

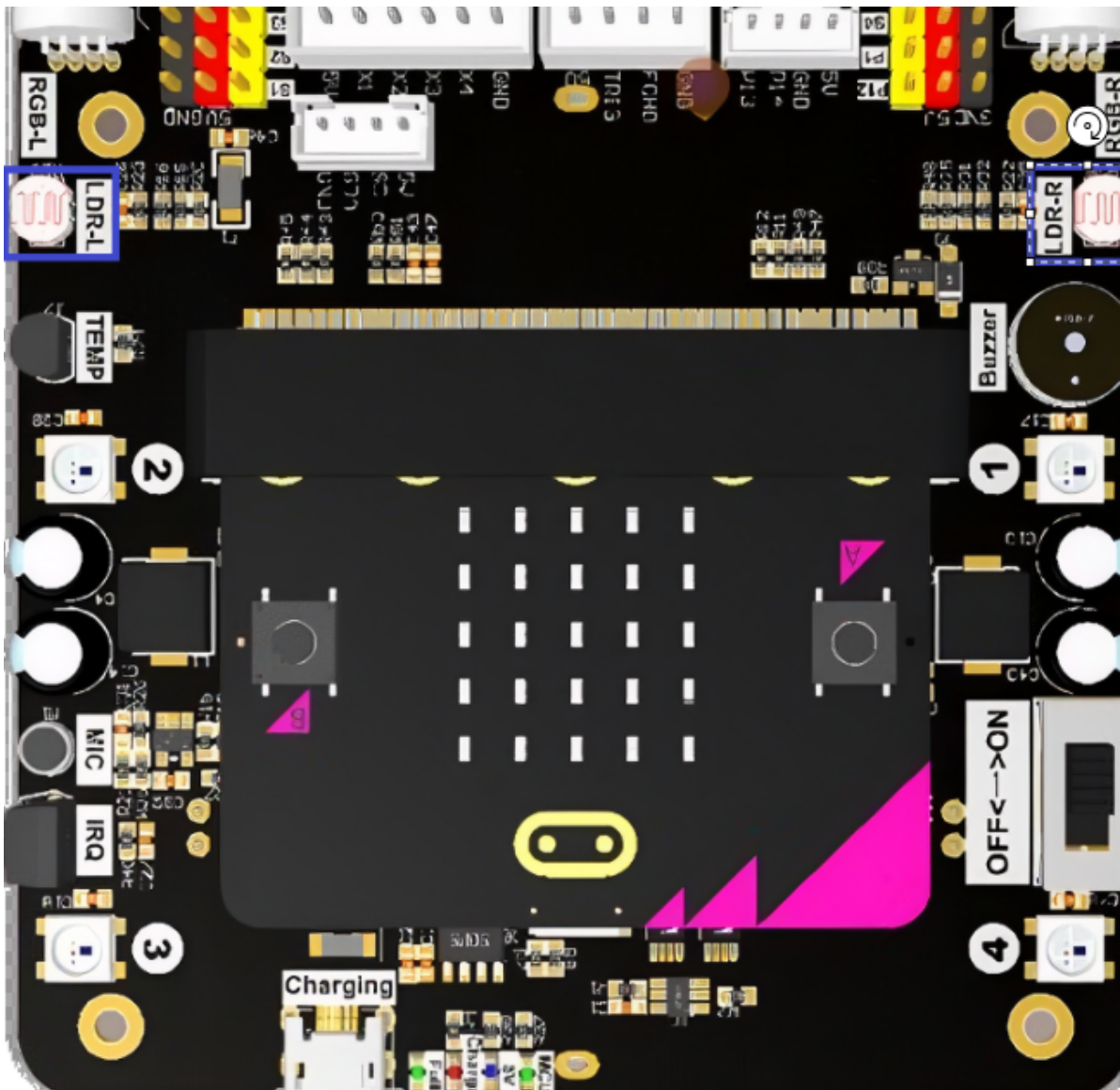
Light Chasing Car

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1. Experiment Purpose
2. Implementation Principle
3. Experiment Steps
4. Experiment Phenomenon

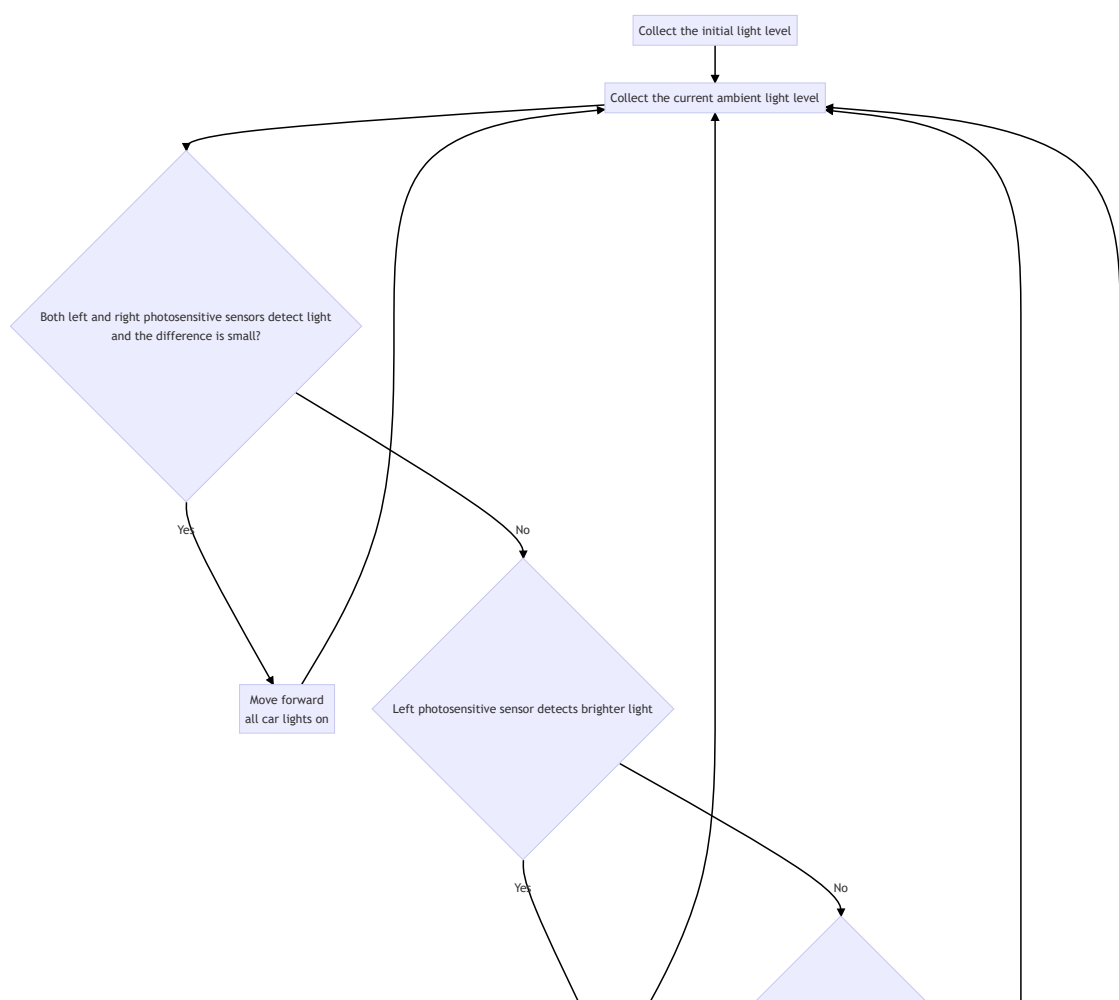
1. Experiment Purpose

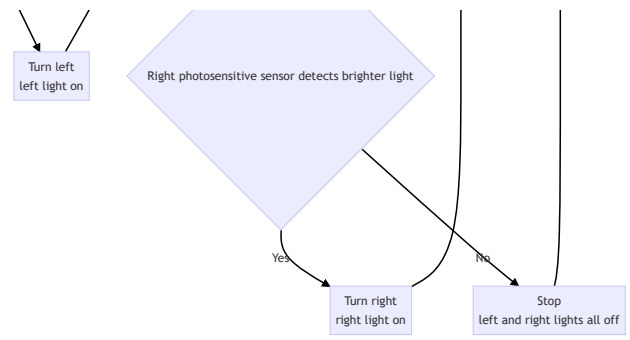
In this section, we will use the photosensitive sensor together with motor and light control to achieve the "light chasing car" effect.



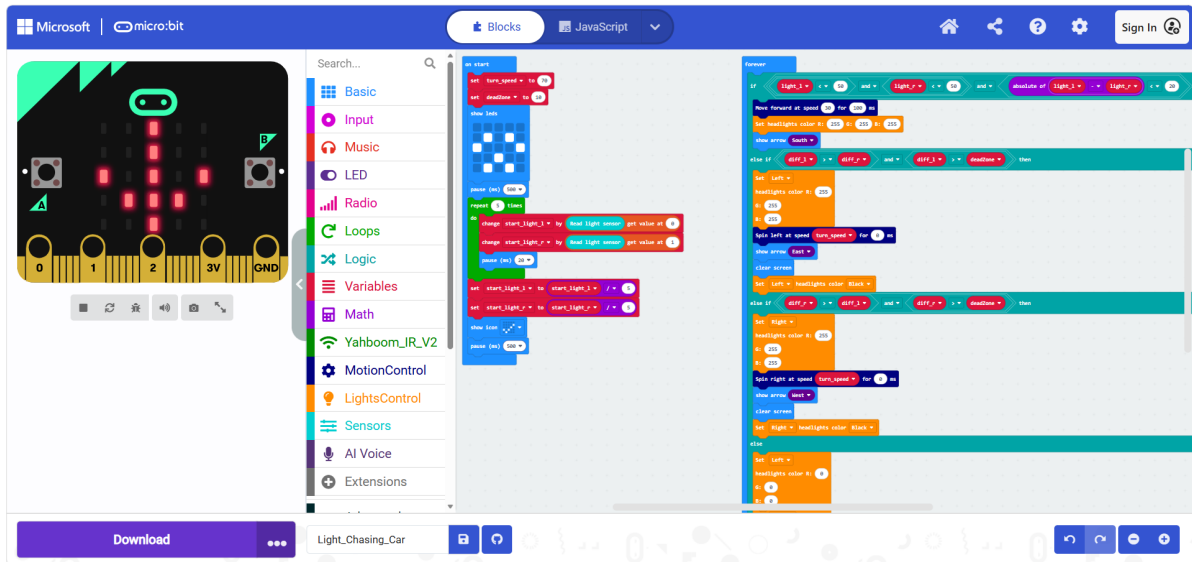
2. Implementation Principle

Thanks to the sensitivity of the photosensitive sensor to light, when the light changes significantly, the read data also changes significantly. The brighter the environment, the smaller the data read by the photosensitive sensor.





The program is as follows.



4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `21.Light_Chasing_Car.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

Since the photosensitive sensor has errors, you can modify the photosensitive sensor thresholds according to the actual situation.

You can shine a flashlight on the photosensitive sensors. Shine it on the left photosensitive sensor and the car will turn left with the left headlight on.



Shine it on the right photosensitive sensor and the car will turn right with the right headlight on.

If both sides detect light at the same time and the intensities are similar, the car will turn on both headlights and drive forward.